

Belgium ELV Material Value Analysis

19 September 2026 · NYU Stern MSBAi Capstone The Global ELV Recycling Gap · material value composition, Belgium 2023

Scope and reading conventions

This analysis decomposes Belgium's end-of-life vehicle stream by material mass and attempts to convert that to value. All figures come from the validated `elv_bronze` path at the 28 April 2026 Eurostat vintage. No API repull was performed.

Every number below carries a tag. The tags are the point of the document: the mass side of this analysis is solid and the value side is not, and the tags are what make that visible at a glance.

Item	Value
GCP project	<code>msbai-capstone-nb4603</code>
Dataset	<code>elv_bronze</code>
Source tables	<code>env_waselvt_api</code> , <code>env_waselv_api</code> , <code>road_eqs_carmot_api</code> , <code>codelists_api</code>
Eurostat vintage	28 April 2026
Reference year	2023
Currency	USD at quoted dates, not converted (see Price assumptions)
Companion note	<code>netherlands_figures_review.md</code>

Tag	Meaning
<code>[DATA]</code>	Measured in <code>elv_bronze</code> , traceable to a table, year and code
<code>[DERIVED]</code>	Computed from <code>[DATA]</code> , with the arithmetic shown
<code>[SOURCED]</code>	External input with a citation and a date
<code>[OPEN]</code>	Required but not obtainable, deliberately left unfilled
<code>[ASSUMPTION]</code>	An assumption, carried forward explicitly

Nothing in this document substitutes a plausible number for a missing one. Where an input could not be sourced it is marked `[OPEN]` and the dependent calculation is not performed.

Headline: four things Eurostat cannot tell you

A mass-share against value-share contrast cannot be computed for Belgium from Eurostat ELV statistics, and the reason is structural rather than a Belgian data quality problem. This is a finding about why a global recycling gap can exist, not a caveat on the analysis.

- 1. Copper cannot be separated from aluminium.** Eurostat reports one combined code, `W191002` "Non-ferrous materials (aluminium, copper, zinc, lead, etc.)". There is no split anywhere in the codelist.

2. **PGM content is not reported.** W1608 "Catalysts" is reported as a gross mass, 93 tonnes in 2023. Platinum, palladium and rhodium content within that mass does not exist as a statistic.
3. **Electronics and semiconductors have no category at all.** There is no code for them anywhere in the ELV waste codelist.
4. **Belgium's EXP code is not used-vehicle exports.** It is "End-of-life vehicles exported", reported in tonnes of waste. Used vehicles shipped abroad for resale leave the fleet through deregistration and appear in no ELV statistic at all.

The first three block the value decomposition. The fourth blocks the Antwerp hypothesis as currently constructed, and is addressed in Export to fleet below.

What the data does support is a mass decomposition that reconciles exactly, a Belgium-specific tonnes-per-vehicle conversion factor, and a 63 percent collapse in ELV intake since 2010 that no variable in this dataset explains.

Rows, coverage and category codes

[DATA] Queried `geo = 'BE'`, all years, both tables. No null observations in either.

Table	Rows (BE)	Year range	Years	Null OBS_VALUE
env_waselvt_api	144	2006 to 2023	18	0
env_waselv_api	1,375	2006 to 2023	18	0

Belgium's series begins at 2006, not 2005. Any panel built alongside countries whose series start in 2005 will be one year offset.

Category codes, resolved against the Eurostat codelist

[DATA] All 17 codes present for Belgium resolve. None unresolved.

Code	Label	Note
ELV	Waste arising only from end-of-life vehicles of type passenger cars (M1), light commercial (N1)	Top-level total
EXP	End-of-life vehicles exported	Tonnes, not counts
DMDP	Waste from dismantling and de-pollution of end-of-life vehicles	Branch total
W1910	Waste arising from shredding of end-of-life vehicles	Branch total
W191001	Ferrous scrap (steel) from shredding	
W191002	Non-ferrous materials (aluminium, copper, zinc, lead, etc.)	Combined, not split
W1910A	Shredder Light Fraction (SLF)	Where unstripped material lands
W1910B	Other materials arising from shredding	
W1601B	End-of-life vehicles: metal components	Ferrous / non-ferrous mix, unresolved
W1601C	End-of-life vehicles: other materials arising from dismantling	Residual, 25.2 percent of stream
W1608	Catalysts	Gross mass only
W1606	Batteries and accumulators	
W160103	End-of-life vehicles: tyres	British spelling in Eurostat's label
W160107	End-of-life vehicles: oil filters	
W160119	End-of-life vehicles: large plastic parts	
W160120	End-of-life vehicles: glass	
LIQ	Liquids (excluding fuel)	

Operation codes, all resolved: **GEN** waste generated, **REU** reuse, **RCY** recycling, **RCV** recovery, **RCV_E** energy recovery (R1), **DSP** disposal.

The Belgium denominator

[DATA] Belgium reports in three units, and the unit matters. **There is exactly one count series: **GEN** in **NR**.** Everything else, exports included, is mass.

Unit	Label	Rows (BE)	What carries it
T	Tonne	1,465	Every material category, and EXP
NR	Number	18	GEN only, in env_waselvt_api
PC	Percentage	36	The reported rate series

ELVs generated, counts and tonnes

Year	Generated (vehicles)	Generated (tonnes)	Implied t/vehicle
2010	170,562	176,446	1.034
2015	107,425	119,054	1.108
2019	134,629	168,810	1.254
2020	110,161	138,468	1.257
2021	103,659	129,979	1.254
2022	81,350	102,334	1.258
2023	63,592	80,190	1.261

[DERIVED] **The count-to-tonne conversion factor is 1.261 t/vehicle for 2023**, computed as 80,190 / 63,592. It is derived from Belgium's own reported data rather than imported, which makes it the defensible factor for this country. It has drifted up from 1.034 in 2010, consistent with vehicles getting heavier, and is stable at about 1.25 to 1.26 across 2019 to 2023.

[DATA] **Belgium's ELV intake has fallen 63 percent from its 2010 peak**, from 170,562 to 63,592 vehicles. Nothing in this dataset explains it.

Treated against generated, and the divergence trap

Year	GEN (t)	RCY (t)	RCV (t)	Gap (t)	Gap as % of GEN
2020	138,468	100,412	106,373	32,095	23.2%
2021	129,979	95,215	100,159	29,820	22.9%
2022	102,334	73,782	78,110	24,224	23.7%
2023	80,190	56,651	60,014	20,176	25.2%

This gap is an artefact, not lost mass. REU , RCV_E and DSP are not populated in `env_waselvt_api` for Belgium in any year. They are populated in the detailed table `env_waselv_api` , and using it closes most of the gap. Anyone working from the totals table alone will conclude Belgium loses a quarter of its ELV mass, which is wrong.

Mass composition, 2023

[DATA] From `env_waselv_api` , mass per category taken as `REU + RCY + RCV_E + DSP` . Ferrous is 45.3 percent of the stream. The combined non-ferrous fraction that would contain all the copper is 6.6 percent, and catalysts, which contain all the PGMs, are 0.12 percent.

Category	Code	Tonnes	Share of stream
Ferrous scrap from shredding	W191001	33,904	45.3%
Other materials from dismantling	W1601C	18,852	25.2%
Shredder Light Fraction	W1910A	6,948	9.3%
Metal components, dismantled	W1601B	4,960	6.6%
Non-ferrous, combined	W191002	4,907	6.6%
Other from shredding	W1910B	1,999	2.7%
Tyres	W160103	1,945	2.6%
Batteries	W1606	564	0.8%
Liquids	LIQ	462	0.6%
Large plastic parts	W160119	121	0.2%
Catalysts	W1608	93	0.12%
Glass	W160120	49	0.07%
Oil filters	W160107	16	0.02%
Total		74,819	100%

The decomposition reconciles exactly. Dismantling (DMDP , 27,061 t) and shredding (W1910 , 47,758 t) sum to 74,819 t. The shredding sub-categories sum to their branch total to the tonne. Treated ELV is reported as 74,954 t, 135 t above the branch sum, a 0.18 percent discrepancy. **The mass side is solid. The weakness is entirely on the value side.**

Two mass figures that should not be believed

Glass at 49 tonnes and large plastic parts at 121 tonnes, against a 74,819 tonne stream, are 0.07 and 0.16 percent. A passenger car is roughly 3 percent glass and 8 to 12 percent plastic by mass, which on this stream would be roughly 2,200 t of glass and 6,000 to 9,000 t of plastic.

[DERIVED] **Belgium is reporting about 2 percent of the glass and about 2 percent of the plastic that must physically be in the stream.** These codes measure separately dismantled material, not material content. The remainder goes into the shredder and arrives in Shredder Light Fraction. This is the clearest single piece of evidence on the stripping question below.

The residual problem

W1601C "other materials arising from dismantling" is **18,852 tonnes, 25.2 percent of the entire stream**, of which 17,898 t is reuse. A quarter of Belgium's ELV mass sits in a category whose label is "other" and whose composition is not reported. [OPEN] No value can be attributed to it without knowing what it contains. Since it is overwhelmingly reuse, it is most likely whole reusable parts, which are the highest-value output of a dismantling operation, but the data does not say so.

The stripping question, unresolved

The evidence points both ways, and the data cannot settle it.

- **Consistent with stripping:** catalysts (93 t), batteries (564 t) and liquids (462 t) appear as separate dismantling outputs, so some selective removal is certainly happening. This is legally mandatory under the ELV Directive for exactly these components.
- **Consistent with no stripping:** glass and plastics appear at roughly 2 percent of their physical presence, so the bulk of those materials goes into the shredder with the shell.

[ASSUMPTION] **Carried forward explicitly:** regulatory stripping only. Catalysts, batteries, liquids, tyres and oil filters removed; copper, aluminium and electronics enter the shredder with the shell. **This is unverified and it materially changes the answer.** If Belgian dismantlers hand-strip harnesses and modules before shredding, recovered value rises and the SLF loss falls. The ELV statistics report treatment routes, not material content, so they cannot resolve it.

Price assumptions and what cannot be computed

Every price used, with source and date. Where a figure could not be sourced it is marked [OPEN] and left unfilled.

#	Input	Value	Source and date	Tag
P1	Ferrous scrap, Europe	340 USD/t	Intratec ferrous scrap prices, January 2026	[SOURCED] stale
P2	Copper, COMEX	~13,900 USD/t	Fastmarkets base metals update, 14 Sep 2026	[SOURCED]
P3	Aluminium	~3,250 USD/t	Trading Economics, mid-Sep 2026	[SOURCED]
P4	Platinum	1,796 USD/oz	Kitco, 19 Sep 2026	[SOURCED]
P5	Palladium	1,314.50 USD/oz	Trading Economics, 18 Sep 2026	[SOURCED]
P6	Rhodium	9,225 USD/oz	Trading Economics, 17 Sep 2026	[SOURCED]
P7	Copper share of W191002	not available	no source	[OPEN]
P8	Aluminium share of W191002	not available	no source	[OPEN]
P9	PGM grams per catalyst tonne	not available	no source	[OPEN]
P10	Semiconductor mass and value	not available	no Eurostat category exists	[OPEN]
P11	Composition of W1601C	not available	not reported, 25.2% of stream	[OPEN]
P12	Plastics, glass, SLF value	~0 or negative	disposal cost, not revenue	[ASSUMPTION]

No USD to EUR conversion has been applied. Mixing a January 2026 scrap price with September 2026 metal prices at a single exchange rate would manufacture false precision. P1 is the weakest input and the largest mass component depends on it; Western European scrap was reported broadly stable through September 2026 (Kallanish), which supports using it, but it should be refreshed before publication.

What can and cannot be computed

[DERIVED] Ferrous value, using P1: $33,904 \text{ t} \times 340 \text{ USD/t} = 11,527,360 \text{ USD}$.

Requested output	Blocked by	Status
Copper value share	P7, no Cu/Al split in W191002	Cannot compute
Aluminium value share	P8, same	Cannot compute
PGM value share	P9, catalyst mass reported but not PGM content	Cannot compute
Semiconductor value share	P10, no category exists	Cannot compute

The mass is concentrated in the cheapest material while the materials that carry most of the value sit in fractions of a few percent. That is the **precondition** for a value inversion. Whether the inversion actually holds in Belgium cannot be established without the [OPEN] inputs above, and must not be asserted from the mass shares alone.

Closing the open inputs: run both methods

The [OPEN] inputs can be closed two different ways, and the two ways answer different questions. Running both is not redundancy. The difference between them is the result.

Method	What you do	What it tells you
Top-down	Apply published material composition per vehicle to Belgium's 63,592 vehicles and 80,190 t	What is physically present in the stream
Bottom-up	Obtain actual recovered tonnages by material from Febelauto or an operator	What was actually recovered

These are not interchangeable. Applying top-down composition figures and labelling the output "recovered value" would be wrong: the number produced is value present, not value captured. **The gap between the two is the recycling gap expressed in materials rather than in compliance percentages,** which is the capstone thesis stated in one number. Bottom-up is also the only route that settles the stripping question above.

Worked reconciliation

[DERIVED] Using the sourced anchors and the mass data above.

Component	Tonnes	Basis
Top-down copper	1,590	25 kg/vehicle x 63,592 vehicles [SOURCED]
Top-down aluminium	6,415	8 percent of 80,190 t [SOURCED]
Non-ferrous present	8,005	sum
Recovered, W191002	4,907	[DATA]
W1601B metal components	4,960	[DATA], Fe / non-Fe split unknown

Because W1601B is an unresolved mix, the answer is a **bracket rather than a point estimate**, which is the honest form:

- If W1601B is entirely ferrous: 3,098 t of non-ferrous is not recovered as non-ferrous, 39 percent of what is present.
- If W1601B is entirely non-ferrous: the gap closes to zero.

The truth sits between these, and **one number from the operator collapses the bracket to a point estimate**. That is a far sharper request than asking for a data extract.

The consistency check that makes this credible

A top-down estimate is only worth reporting if the implied missing mass has somewhere physical to go. It does.

unrecovered non-ferrous (upper bound)	3,098 t
Shredder Light Fraction (W1910A)	6,948 t
	$3,098 / 6,948 = 45\%$

The missing non-ferrous fits inside SLF with room to spare. **If the method had implied 20,000 t of unaccounted copper it would be broken and should be discarded.** It does not, so the estimate survives its own sanity check. Report this check alongside the bracket: it is what distinguishes an estimate from a guess.

The vintage correction, which is not optional

[ASSUMPTION] **The most likely challenge.** The 8 percent aluminium anchor is a current-fleet figure. Vehicles scrapped in Belgium in 2023 were built around **2005 to 2010**, since average EU scrappage age is roughly 15 years. Aluminium content in new cars has risen substantially over that period, so applying a 2023 new-car composition to a 2008-build vehicle **overstates aluminium, plausibly by up to a third.**

The correction is to use composition **at build year**. This is precisely what a material composition *trends* report exists to provide. **Copper travels better across vintages than aluminium does**, since wiring harness mass has been comparatively stable, so the aluminium figure carries most of the vintage risk.

Where the missing inputs come from

Open input	Route	Source and status
P7, P8 copper and aluminium share	Top-down	JRC126564, "Material composition trends in vehicles", European Commission Joint Research Centre. Lead, not verified: the domain is blocked from the analysis sandbox, so contents could not be confirmed
P8 aluminium	Top-down	European Aluminium, <i>Aluminium Content in Passenger Vehicles (Europe)</i> . Roughly 123 kg castings per vehicle, roughly 80 kg in powertrain
P9 PGM grams per tonne	Both	Johnson Matthey PGM Market Report, free and annual, the industry reference. Market-level recovery; per-converter loading may need a teardown study on top
P10 semiconductors	Top-down only	No statistical source exists. Teardown and academic literature only. May stay [OPEN]
P7, P9 and the W1601B split	Bottom-up	Febelauto , Belgium's ELV compliance scheme. See appendix

Why Febelauto is the right counterparty, evidenced rather than assumed. Febelauto reported **81,350 vehicles collected in 2022**, which matches the Eurostat GEN count for Belgium in 2022 in this dataset **exactly**. That identity establishes the reporting chain as ATFs to Febelauto to Eurostat, which means Febelauto holds the granularity Eurostat aggregates away before publication. They operate a network of over 100 authorised treatment facilities.

Incidental finding for the scrap price workstream. That "over 100 ATFs" figure is an operator count for Belgium. The scrap price case needs an operator concentration measure and found permitted storage capacity to be unharmonised across the EU. Compliance schemes may hold ATF counts per country that Eurostat SBS does not publish at 4-digit NACE. Worth checking before settling for the SBS route.

Export to fleet, and the Antwerp hypothesis

[DATA] Belgium passenger car fleet from `road_eqs_carmot_api`, `mot_nrg=TOTAL`, `engine=TOTAL`, unit NR : **6,047,551 in 2023** (5,955,127 in 2022; 5,926,009 in 2021). Same Eurostat vintage, already in `elv_bronze`, no repull.

[DATA] Belgium's EXP is reported in tonnes and means "End-of-life vehicles exported". The 2023 values are 5,371 t on a GEN basis and 5,020 t across treatment operations.

The measurement problem, stated plainly

Two different populations are easily conflated here, and any export-to-fleet ratio depends entirely on which one is meant.

	Used-vehicle exports	Eurostat EXP
Unit	vehicles (count)	tonnes (mass)
Population	used vehicles exported for resale	end-of-life vehicles exported for treatment
Leaves as	a roadworthy vehicle	waste
In ELV statistics?	No	Yes

A used vehicle driven onto a ship at Antwerp is not an end-of-life vehicle and never enters `env_waselv_api`. It leaves the fleet through deregistration and appears in no ELV statistic at all. This is the border where the regulatory statistics stop, and it is the whole of the Antwerp hypothesis.

Computing the ratio anyway

[DERIVED] Using the 1.261 t/vehicle factor: $5,371 \text{ t} / 1.261 = 4,259 \text{ vehicle-equivalents}$.

Construction	Belgium 2023
EXP vehicle-equivalents / registered fleet	$4,259 / 6,047,551 = \mathbf{0.07\%}$
EXP tonnes / generated tonnes	$5,371 / 80,190 = \mathbf{6.7\%}$
EXP vehicle-equivalents / ELVs generated	$4,259 / 63,592 = \mathbf{6.7\%}$

Which of these is the right ratio depends on what it is being compared against. A ratio built on the registered fleet and one built on annual ELV volume differ here by a factor of roughly 100. They are not

interchangeable, and a cross-country comparison that mixes the two will report a difference that is entirely definitional.

The Antwerp hypothesis, tested

The proposition was that Belgium's used-vehicle export volume through Antwerp is large relative to its domestic fleet, and that this should show up as a high export-to-fleet ratio.

The ELV statistics cannot test it. EXP measures end-of-life vehicles exported for treatment, not used vehicles exported for resale. The Antwerp flow leaves as roadworthy vehicles and is absent from these tables entirely.

This is a measurement finding, not a null result. The hypothesis is not disproved; it is untestable on this data and needs a vehicle-registration or trade source instead.

Reporting history

[DATA] What the data does support: Belgium's ELV exports have collapsed, from 31.4 percent of generated tonnage in 2006 to 6.7 percent in 2023.

Year	EXP (t)	Generated (t)	EXP as % of generated
2006	41,079	131,030	31.4%
2010	37,031	176,446	21.0%
2014	8,630	138,703	6.2%
2018	26,890	177,439	15.2%
2021	12,140	129,979	9.3%
2022	6,945	102,334	6.8%
2023	5,371	80,190	6.7%

[DERIVED] Note the 2014 to 2018 reversal. The ratio fell to 6.2 percent in 2014, recovered to 15.2 percent by 2018, then fell again. A monotonic decline would suggest structural change; this looks more like a reporting or policy discontinuity around 2014 to 2016. Any trend claim on EU export reporting should be checked against that break first.

Open items, ranked by what they block

#	Item	What it blocks, and how to close it
1	Ferrous / non-ferrous split of <code>W1601B</code>	Collapses the recovery bracket from "0 to 3,098 t" to a point estimate. One number from Febelauto
2	Per-vehicle catalyst PGM loading	All PGM value. Johnson Matthey, or an operator assay
3	Copper share of <code>W191002</code>	All copper value. JRC composition data, or an operator
4	Stripping behaviour before shredding	Whether recovered value is understated. Only obtainable from an operator
5	Composition of <code>W1601C</code>	25.2 percent of stream mass carries no attributable value
6	Refreshed European ferrous scrap price	P1 is from January 2026 and the largest mass component depends on it
7	Semiconductor mass per vehicle	No statistical source exists; may remain permanently open

Comparator data issues raised by this analysis have moved to the companion note, `netherlands_figures_review.md`, so they reach the owner of that run rather than sitting inside a Belgium deliverable.

Likely questions

- "Your non-ferrous number contains both copper and aluminium. How can you claim anything about copper?"** I cannot, and I say so. `W191002` is a single code covering aluminium, copper, zinc and lead. The split is [OPEN].
- "You report 93 tonnes of catalysts. What is the PGM content?"** Not reported. Catalyst mass is gross. Grams per tonne is [OPEN] and is one of only two numbers needed to complete the value picture.
- "Why is Belgium reporting 49 tonnes of glass?"** Because the code measures separately dismantled glass, not glass content. Physical glass is roughly 2,200 t, so about 98 percent goes to the shredder. This is the clearest evidence on the stripping question.
- "A quarter of your mass is in a category called 'other'. What is in it?"** Unknown. `W1601C`, 18,852 t, overwhelmingly reuse, most likely whole reusable parts. [OPEN].
- "Your generated and recovery totals differ by 25 percent. Where did the mass go?"** Nowhere. `REU`, `RCV_E` and `DSP` are unpopulated in the totals table and populated in the detailed table. This trips up anyone using `env_waselvt_api` alone.
- "Does the Antwerp export story show up in the numbers?"** Not in these numbers, and it cannot. `EXP` is ELV waste exported for treatment; used vehicles exported for resale are absent from ELV statistics entirely. Untestable here rather than disproved.
- "Do you know whether high-value materials are stripped before shredding?"** No. Regulatory depollution is happening and glass and plastic are not being separated. "Regulatory stripping only" is carried forward as an explicit assumption that materially changes the value result.
- "Why is your scrap price from January when your metal prices are from September?"** Because that is the most recent European ferrous scrap price I could source. It is the weakest input and the largest mass component depends on it.

9. "Belgium's ELV intake fell 63 percent since 2010. Does that not dominate everything?" It may. 170,562 to 63,592 vehicles is not explained within this dataset, and any value total scales directly with it.

Provenance

Figure	Source	Tag
Row counts, year coverage	<code>env_waselvt_api</code> , <code>env_waselv_api</code> , <code>geo='BE'</code>	[DATA]
Category labels	<code>codelists_api</code> , Eurostat DSD	[DATA]
ELVs generated, counts and tonnes	<code>env_waselvt_api</code> , <code>wst_oper='GEN'</code> , units NR and T	[DATA]
1.261 t/vehicle	80,190 / 63,592, both [DATA]	[DERIVED]
2023 mass decomposition	<code>env_waselv_api</code> , 2023, REU+RCY+RCV_E+DSP	[DATA]
Fleet 6,047,551	<code>road_eqs_carmot_api</code> , TOTAL/TOTAL, NR	[DATA]
<code>EXP</code> series	<code>env_waselv_api</code> , <code>waste='EXP'</code>	[DATA]
All prices	Price assumptions table, each with source and date	[SOURCED] / [OPEN]
Febelauto 81,350 for 2022	Recycling International; matches Eurostat GEN exactly	[SOURCED]
Stripping behaviour	not resolvable from Eurostat	[ASSUMPTION]

Eurostat vintage 28 April 2026 throughout. No API repull was performed. All Belgium figures come from the validated `elv_bronze` tables.

Appendix: data request to Febelauto

Three specific numbers, not a data extract. Each names the gap it closes. A narrow, justified ask is far more likely to be answered than a general one.

Ask	Number wanted	Why, and what it closes
1	Ferrous against non-ferrous split of dismantled metal components (<code>W1601B</code> , 4,960 t in 2023)	Currently the single number preventing a point estimate. Collapses the bracket from "between 0 and 3,098 tonnes" to one figure
2	Platinum, palladium and rhodium per tonne of catalyst (<code>W1608</code> , 93 t in 2023, gross mass only). Grams per tonne or total grams, either works	Without it no PGM value can be computed at all
3	Whether wiring harnesses and electronic modules are removed before shredding. A yes or no with an approximate share is enough	Settles the assumption carried forward above, which materially changes the recovered value estimate
4 (optional)	Authorised treatment facilities per year	Operator concentration series for the scrap price workstream, which Eurostat SBS may suppress at 4-digit NACE 38.31

Useful context for the approach. Their published figure of 81,350 vehicles collected in 2022 matches the Eurostat `GEN` count for Belgium exactly. Saying so shows the request comes from someone who has already

reconciled the published data and is asking only for what sits beneath it.

If Febelauto cannot share operator-level data, Asks 1 and 2 are also obtainable from any single Belgian shredder operator, and Ask 2 from published recycling-industry assay coefficients. Ask 3 has no documentary substitute and would remain [OPEN].

Before publication

- Refresh the ferrous scrap price (P1, January 2026) and state its date.
- Apply the build-year vintage correction to the aluminium anchor, or state plainly that the top-down figure is uncorrected.
- Open JRC126564 and verify it carries the composition split. It is currently a lead, not a confirmed source.
- Report the bracket together with its SLF consistency check. The check is what makes the bracket credible.
- Confirm the comparator construction in the companion Netherlands note before the two countries appear on one slide.